

Homework #1

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1 Insertion Sort

1.1 Operations of Insertion Sort

3 <u>41</u> 52 26 38 57 9 49	1 Comparison(s).
3 41 <u>52</u> 26 38 57 9 49	1 Comparison(s).
3 41 52 <u>26</u> 38 57 9 49	3 Comparison(s).
3 26 41 52 <u>38</u> 57 9 49	3 Comparison(s).
3 26 38 41 52 <u>57</u> 9 49	1 Comparison(s).
3 26 38 41 52 57 <u>9</u> 49	6 Comparison(s).
3 9 26 38 41 52 57 <u>49</u>	3 Comparison(s).
3 9 26 38 41 49 52 57	0 Comparison(s).

1.2 Number of Comparisons

18 total operations.

2 Question 2

B	A	O	o	Ω	ω	Θ
2^n	$2^{\frac{n}{2}}$	no	no	yes	yes	no
$n^{\lg c}$	$c^{\lg n}$	yes	no	yes	no	yes

2.1 Justification

To determine larger asymptotic growth, take the limit of one function over the other. Arbitrarily choosing 2^n for the numerator, we see that:

$$\lim_{n \rightarrow \infty} \frac{2^n}{2^{\frac{n}{2}}} = \lim_{n \rightarrow \infty} 2^{\frac{n}{2}} = \infty \quad (1)$$

Alternatively, if we chose 2^n as the denominator, we notice that $\lim_{n \rightarrow \infty} \frac{2^{\frac{n}{2}}}{2^n} = 0$, so we know that the condition $2^n > 2^{\frac{n}{2}}$ holds, and for certain conditions the functions are equal (take $n = 0, 2^0 = 2^{\frac{0}{2}}$).
 $\therefore 2^n$ is $\omega(2^{\frac{n}{2}})$ and $\Omega(2^{\frac{n}{2}})$. QED.

2.2 Justification II

By the properties of logarithms, we will show that $n^{\lg c} = c^{\lg n}$.

$$n^{\lg c} = c^{\lg n} \quad (2)$$

$$= c^{\log_2 n} \quad (3)$$

$$= c^{\frac{\ln n}{\ln 2}} \quad (4)$$

$$= e^{\ln c \times \frac{\ln n}{\ln 2}} \quad (5)$$

$$= e^{\frac{\ln c}{\ln 2} \times \ln n} \quad (6)$$

$$= e^{\lg c \ln n} \quad (7)$$

$$= n^{\lg c} \quad (8)$$

$\therefore n^{\lg c}$ is $O(c^{\lg n})$, $\Omega(c^{\lg n})$ and $\Theta(c^{\lg n})$. QED.

3 Big-O Implies Big-Ω

Theorem: Let $f(n)$ and $g(n)$ be asymptotically positive functions, $O(g(n))$ be the set $\{f(n) : \exists c, n_0 \in \mathbb{R}^+, \forall n, n_0 \in \mathbb{R}, 0 \leq f(n) \leq c g(n) \wedge n > n_0\}$, and $\Omega(g(n))$ be the set $\{f(n) : \exists c, n_0 \in \mathbb{R}^+, \forall n, n_0 \in \mathbb{R}^+, 0 \leq c g(n) \leq f(n) \wedge n > n_0\}$. Then,

$$f(n) = O(g(n)) \Rightarrow g(n) = \Omega(f(n)) \quad (9)$$

[We will prove so by contradiction.]

Proof: Suppose not. That is, suppose

$$\sim [f(n) = O(g(n)) \Rightarrow g(n) = \Omega(f(n))] \quad (10)$$

$$f(n) = O(g(n)) \wedge \sim [g(n) = \Omega(f(n))] \quad (11)$$

Using the formal definition of $\Omega(g(n))$, we can see the negation is as follows:

$$\sim \Omega(g(n)) = \{f(n) : \forall c, n_0 \in \mathbb{R}^+, \exists n, n_0 \in \mathbb{R}^+, 0 > c g(n) > f(n) \wedge n \leq n_0\} \quad (12)$$

This is a contradiction, for there are no c, n_0 such that $0 \leq c g(n) \leq f(n) \wedge 0 > c f(n) > g(n)$ because c, n_0 are defined as *positive* constants, and $f(n), g(n)$ are defined as *positive* functions. Because no such positive function and positive constants exists to satisfy

$$0 > cf(n) > g(n) \quad (13)$$

This has led us to a contradiction. QED.

4 Asymptotic Proofs

4.1 Proof I

Assuming $n > 1$,

$$n^2 \leq 20n^2 + 2n + 5 \quad (14)$$

$$\leq 20n^2 + 2n^2 + 5 \quad (15)$$

$$\leq 27n^2 \quad (16)$$

$\therefore$ For $C = 27, n_0 = 1, 20n^2 + 2n + 5 = O(n^2)$. QED.

4.2 Proof II

Assume $C = 1, n_0 = 1$. $\therefore$ This satisfies the condition that c and n_0 are positive constants such that $0 \leq Cn^2 \leq 5n^2 - 15n + 100 \forall n \geq n_0$. QED.

4.3 Proof III

4.3.1 Lower Bound

Assume $C = 1, n_0 = 1$. $\therefore$ This satisfies the condition that c and n_0 are positive constants such that $0 \leq Cn^2 \leq 5n^2 + 2n \forall n \geq n_0$. QED.

4.3.2 Upper Bound

Assuming $n > 1$,

$$n^2 \leq 5n^2 + 2 \quad (17)$$

$$\leq 5n^2 + 2n^2 \quad (18)$$

$$\leq 7n^2 \quad (19)$$

$\therefore C = 7, n_0 = 1, 5n^2 + 2n = \Theta(n^2)$. QED.

4.4 Proof IV

To prove that $5n + 7 = o(n^2)$ we must show that $\exists c, n_0 \in \mathbb{R}^+, 0 \leq f(n) < Cg(n) \wedge n > n_0$. We do so by showing that $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$. [*with the help of L'Hôpital's Rule*]

$$\lim_{n \rightarrow \infty} \frac{5n + 1}{n^2} \tag{20}$$

$$= \lim_{n \rightarrow \infty} \frac{5}{2n} \tag{21}$$

$$= 0 \tag{22}$$

QED.